Lesson 1: Volume of Prisms and Cylinders

Counting the cubes that fill a space, using V = (area of base) × height for prisms and cylinders, changing cubic units, and ordering concrete, soil and gravel.

1. What Is Volume?

The **volume** of an object is the amount of **space** it takes up. For a box, a tank or a truck, it also tells you how much fits inside. We measure volume by counting how many **cubes** would fill the space.

- A **cubic centimetre** (1 cm^3) is a cube 1 cm long, 1 cm wide and 1 cm high, about the size of a sugar cube.
- A **cubic inch** (1 in^3) is a cube 1 in on each edge, about the size of a small ice cube.
- A **cubic metre** (1 m^3) is a cube 1 m on each edge, about as tall as a kitchen counter. A **cubic foot** (ft^3) and a **cubic yard** (yd^3) are also used for big jobs.

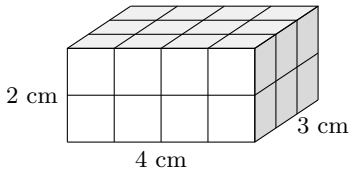
The small 3 means **cubed**: three lengths were multiplied. Say "cubic centimetres" for cm^3.

A **rectangular prism** is a box shape with 6 rectangle faces. To count its cubes, count the cubes in the **bottom layer**, then multiply by the number of **layers**.

Rectangular prism: $V = l \times w \times h$ (length × width × height)
Cubes in the bottom layer $= l \times w =$ **area of the base**. Number of layers $= h$. So $V =$ (area of base) × height.

Watch out. All three lengths must be in the **same unit** before you multiply. Volume has a **cubed** unit (cm^3, ft^3). Area has a squared unit (cm^2), and length has no small number (cm).

Example 1



This box is built from 1 cm cubes.

(a) How many cubes are in the bottom layer? _____

(b) How many layers are there? _____

(c) What is the volume of the box? _____

(d) Check with the formula $V = l \times w \times h$. _____

Example 2

Find the volume of each rectangular prism. Round to the nearest hundredth where needed.

Object	Length × width × height	Work	Volume
(a) Cereal box	19 cm × 7 cm × 28 cm		
(b) Shoe box	13 in × 7.5 in × 5 in		
(c) Shipping container	12 m × 2.4 m × 2.6 m		
(d) Wooden storage box	$2\frac{1}{2}$ ft × $1\frac{1}{2}$ ft × $1\frac{1}{4}$ ft		

Calculator. For fractions, change to decimals first: $\frac{1}{2} = 0.5$, $\frac{1}{4} = 0.25$, $\frac{3}{4} = 0.75$. So $2\frac{1}{2}$ ft is 2.5 ft. (You can also enter $2\frac{1}{2}$ as 2 [A b/c] 1 [A b/c] 2.)

2. Volume of Any Prism

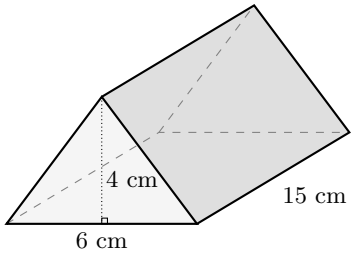
A **prism** is a 3-D shape with two matching ends, called the **bases**, joined by flat rectangle sides. A prism is the same shape all the way through, like a loaf of bread: every slice is the same. A prism is named for its base: a **triangular prism** has triangle bases.

The idea from Section 1 works for every prism: find the area of the base, then multiply by how far the prism goes.

Any prism: $V =$ (area of base) $\times$ height of the prism
Step 1: Find the area of the base (the end shape) with an area formula. Triangle: $A = \frac{1}{2}bh$.
Step 2: Multiply by the height of the prism (the distance from one base to the other; for a prism lying down, its length).

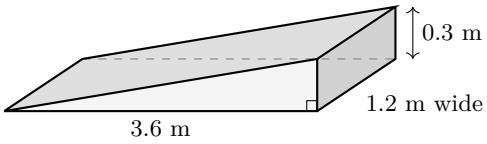
Watch out. A triangular prism has **two** "heights": the height of the triangle (use it for the base area) and the length of the prism. Use each one once. And the base is the **end shape**, not always the bottom.

Example 3



A chocolate bar comes in a box shaped like a triangular prism. Each triangle end has a base of 6 cm and a height of 4 cm. The box is 15 cm long. Find the volume of the box.

Example 4



A concrete wheelchair ramp is a triangular prism lying on its side. The triangle (the side of the ramp) is 3.6 m along the ground and 0.3 m high. The ramp is 1.2 m wide.

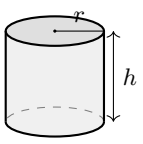
(a) How much concrete is in the ramp?

(b) Concrete costs \$245.00 per m^3. What does the concrete cost?

3. Volume of a Cylinder

A **cylinder** has two matching **circle** bases joined by a curved side, like a can. A cylinder is the same all the way up, so the prism idea still works: area of base $\times$ height. The base is a circle, with area πr^2.

Cylinder: $V = \pi r^2 h$ (area of the circle base $\times$ height)
r is the **radius** (centre to edge). If you know the **diameter** (all the way across), $r = d \div 2$.
Estimate with $\pi \approx 3$ and rounded numbers to check your answer.



Calculator. For $r = 3.7$ and $h = 10.2$: $\boxed{\pi}$ $\boxed{\times}$ 3.7 $\boxed{x^2}$ $\boxed{\times}$ 10.2 $\boxed{\text{ENTER}}$ shows 438.68... The $\boxed{x^2}$ key squares only the number just before it.

Watch out. r^2 means $r \times r$, not $r \times 2$. And square the **radius**, not the diameter: halve the diameter first.

Example 5

(a) A soup can is 7.4 cm across (diameter) and 10.2 cm tall. Estimate its volume, then calculate it to the nearest tenth.

(b) A round hay bale is a cylinder 5 ft across and 4 ft wide (its "height" when it stands on end). Find its volume to the nearest hundredth.

(c) A farmer has 60 of these bales. What is their total volume, to the nearest cubic foot?

Example 6

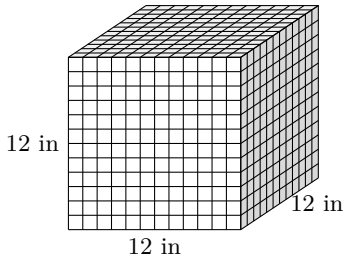
Marisol has a cake recipe for one 9 in round pan that is 2 in deep. She only owns an 8 in by 8 in square pan, also 2 in deep.

(a) Find the volume of each pan, to the nearest hundredth. Will the batter fit in the square pan?

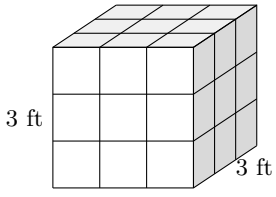
(b) A 9 in by 13 in pan, 2 in deep, holds how many round pans of batter? Round to the nearest tenth.

4. Converting Cubic Units

How many cubic inches are in a cubic foot? A foot is 12 in, so a 1 ft cube is 12 in long, 12 in wide and 12 in high. Its bottom layer has $12 \times 12 = 144$ cubes, and there are 12 layers.



$1\ \text{ft}^3 = 12 \times 12 \times 12$
$= 1728\ \text{in}^3$



$1\ \text{yd}^3 = 3 \times 3 \times 3$
$= 27\ \text{ft}^3$

Within a system (exact):

$1\ \text{ft}^3 = 12 \times 12 \times 12 = 1728\ \text{in}^3$

$1\ \text{yd}^3 = 3 \times 3 \times 3 = 27\ \text{ft}^3$

$1\ \text{m}^3 = 100 \times 100 \times 100 = 1\,000\,000\ \text{cm}^3$

Between SI and imperial (rounded):

$1\ \text{m}^3 \approx 1.31\ \text{yd}^3$ $\quad 1\ \text{yd}^3 \approx 0.765\ \text{m}^3$

$1\ \text{m}^3 \approx 35.3\ \text{ft}^3$

Big unit → small unit: **multiply**. Small unit → big unit: **divide**.

Watch out. 1 ft = 12 in, but 1 ft^3 is **not** 12 in^3. The 12 is used three times (length, width and height). In the same way, 1 m^3 is not 100 cm^3.

Example 7

Convert each volume.

Change	Work	Answer
(a) 3.5 ft^3 to in^3		
(b) 5184 in^3 to ft^3		
(c) 4 yd^3 to ft^3		
(d) 81 ft^3 to yd^3		
(e) 0.25 m^3 to cm^3		
(f) 45 000 cm^3 to m^3		

Another way: change the lengths first. You can also change each length to the unit you want, **before** you find the volume. This is often easier, and it works between SI and imperial too (1 in = 2.54 cm, 1 ft = 30.48 cm, 1 yd = 0.9144 m).

Example 8

A toy chest is 30 in long, 18 in wide and 16 in high. Find its volume in cubic feet two ways.

(a) Find the volume in in^3, then change it to ft^3.

(b) Change each length to feet first ($\div 12$), then find the volume.

5. Volume on the Job: Concrete, Soil and Gravel

Concrete, gravel, sand, topsoil and mulch are sold by volume: by the **cubic yard** or the **cubic metre**. On a job site, "a yard of concrete" means 1 yd^3. The thickness of a slab or a layer is usually small (inches or centimetres), so change it to feet or metres first: $4\ \text{in} = 4 \div 12\ \text{ft}$, $10\ \text{cm} = 0.1\ \text{m}$.

Ordering material:

Step 1: Write all the lengths in the same unit (feet or metres). Step 2: Find the volume.
Step 3: Change to the unit the supplier sells ($\text{ft}^3 \div 27 = \text{yd}^3$). Step 4: **Round up** to an amount you can buy.
Step 5: Find the cost.

Watch out. Always round **up** when ordering. If you round down, the job comes up short, and a second small delivery costs extra.

Example 9

Tyrell is pouring a concrete patio 12 ft long and 10 ft wide. The slab will be 4 in thick. Concrete is sold in half cubic yards (1, 1.5, 2, ...) and costs \$215.00 per yd^3.

(a) Find the volume of the slab in ft^3.

(b) Change it to yd^3. How much should Tyrell order?

(c) What will the concrete cost?

Example 10

A farm driveway is 15 m long and 3.5 m wide. It needs a layer of gravel 10 cm deep. The gravel pit sells only whole cubic yards, at \$48.00 per yd^3. Use 1 $m^3 \approx 1.31$ yd^3.

(a) Find the volume of gravel in m^3.

(b) How many cubic yards should be ordered, and what will they cost?

Working backwards. If you know the volume and the base, you can find the height (or thickness, or depth). Undo the multiplication by dividing.

$V = (\text{area of base}) \times h$ so $h = V \div (\text{area of base})$ For a box: $h = V \div (l \times w)$

Example 11

Mia has 0.9 m^3 of concrete left in the truck. She uses it all to pour a slab for a shed, 3 m by 2.5 m. How thick will the slab be, in centimetres? Check your answer.

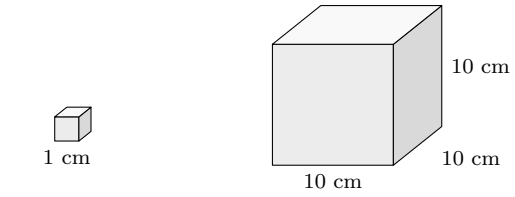
Lesson 2: Capacity: Litres, Gallons, and US vs British Units

Measuring how much a container holds in millilitres, litres, cups and gallons, and why a US gallon is not the same as a British gallon.

1. Capacity and Volume: Millilitres, Litres and Cubic Centimetres

Volume is the amount of space an object takes up. **Capacity** is the amount a container can **hold**, usually of a liquid like water, milk or gas. A jug's capacity is the volume of the space inside it.

In the SI (metric) system, capacity is measured in **litres** (L) and **millilitres** (mL). "Milli" means one thousandth, so 1000 mL = 1 L. Capacity and volume are linked: a cube 1 cm on each side holds exactly 1 mL, and a cube 10 cm on each side holds exactly 1 L.



$1\text{ cm}^3 = 1\text{ mL}$ $10 \times 10 \times 10 = 1000\text{ cm}^3 = 1\text{ L}$

$1\text{ mL} = 1\text{ cm}^3$	$1\text{ L} = 1000\text{ mL} = 1000\text{ cm}^3$	$1\text{ m}^3 = 1000\text{ L}$	
L → mL: $\times 1000$	mL → L: $\div 1000$	m^3 → L: $\times 1000$	L → m^3: $\div 1000$

Referents. A teaspoon holds about 5 mL. A can of pop holds 355 mL. A carton of milk holds 1 L or 2 L, and a big jug 4 L. A car's gas tank holds about 50–70 L. A bathtub holds about 150–200 L.

Example 1

Convert each measurement.

	Work	Answer		Work	Answer
(a) 2.5 L = _____ mL			(b) 750 mL = _____ L		
(c) 340 cm^3 = _____ mL			(d) 4.2 L = _____ cm^3		
(e) 2.6 m^3 = _____ L			(f) 1800 L = _____ m^3		

How much does a container hold? Find the volume of the inside (Lesson 1), then change the cubic units to litres.

Capacity: find the volume inside in cm^3 (then $\div 1000$ for L) or in m^3 (then $\times 1000$ for L).

Example 2

Nadia's fish tank is a rectangular prism. Inside, it is 60 cm long, 30 cm wide and 40 cm high.

(a) Find the volume of the tank in cm^3.

(b) What is the capacity of the tank in litres?

(c) Nadia fills it with water to 4 cm below the top. How many litres of water is that?

Example 3

A rancher near Brooks has a round stock tank for her cattle. It is a cylinder with a diameter of 2.4 m and a height of 0.6 m.

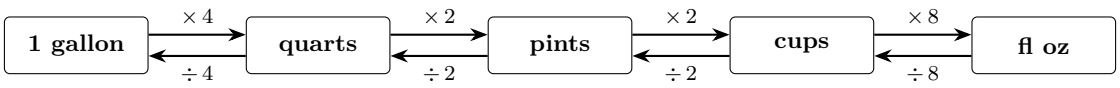
(a) Find the volume of the tank in m^3. Round to three decimal places.

(b) How many litres of water does the tank hold? Round to the nearest litre.

Watch out. The formula uses the **radius**, not the diameter. Halve the diameter first.

2. Imperial Capacity: Fluid Ounces, Cups, Pints, Quarts and Gallons

Canada uses litres, but you still see imperial capacity units on US recipes, on drinks bottled for the US, at gas stations in the US, and on older tools and machines. The imperial units, from smallest to largest, are the **fluid ounce** (fl oz), the **cup**, the **pint** (pt), the **quart** (qt) and the **gallon** (gal). The US units fit together like this:



bigger unit → smaller unit: **multiply** smaller unit → bigger unit: **divide**

US units: 1 gal = 4 qt 1 qt = 2 pt 1 pt = 2 cups 1 cup = 8 fl oz
So 1 gal = 4 qt = 8 pt = 16 cups = 128 fl oz.

Watch out. A **fluid** ounce measures capacity (how much liquid). An ounce (oz) on a scale measures mass. They are different units (mass is Lesson 4).

Example 4

(a) Convert each US measurement.

	Work	Answer
3 gal = ___ qt		
5 pt = ___ cups		
24 fl oz = ___ cups		
6 qt = ___ gal		
2 gal = ___ pt		
1 gal = ___ fl oz		

(b) At a baseball tournament in Montana, the team's concession stand makes 3 US gallons of lemonade. It sells the lemonade in 12 fl oz cups for $2.50 each. How many cups can it fill, and how much money will it take in if it sells them all?

3. US Gallons vs British Gallons

Here is the tricky part: the word "gallon" means **two different sizes**. The **US gallon** is used in the United States. The **British gallon** (also called the **imperial gallon**) was used in Britain and in Canada before Canada changed to metric in the 1970s. You still see it on older Canadian farm equipment, boats and recipes.

Both systems say 1 gallon = 4 quarts = 8 pints. But a British pint holds 20 British fluid ounces, while a US pint holds

16 US fluid ounces. So the British pint, quart and gallon are all about **20% bigger** than the US ones.

	US	**British (imperial)**
1 gallon	≈ 3.785 L	≈ 4.546 L
1 pint	16 US fl oz $\approx$ 473 mL	20 British fl oz $\approx$ 568 mL
1 fluid ounce	≈ 29.57 mL	≈ 28.4 mL

Watch out. When a question says "gallon", check which country. In this booklet, a gallon is a **US gallon** unless it says British. Gas pumps in the US sell by the US gallon.

Example 5

(a) How many litres are in 5 US gallons? In 5 British gallons? How many more litres are in 5 British gallons? Round to the nearest hundredth.

(b) A restaurant in London, England, serves a pint of juice (568 mL). A restaurant in Seattle serves a US pint. How many millilitres is a US pint? How many more millilitres are in the British pint?

(c) Show that a British gallon is about 20% bigger than a US gallon. (Hint: percent increase from Unit 1.)

4. Converting Between SI and Imperial Capacity

To change a capacity between the two systems, multiply or divide by the matching number. Write the units in each step so you can see which way to go. A gallon is **bigger** than a litre (1 US gal ≈ 3.785 L), and a fluid ounce is **bigger** than a millilitre (1 US fl oz ≈ 29.57 mL). So going **to** L or mL you get more units: **multiply**. Going **back** to gallons or fluid ounces you get fewer units: **divide**.

US gal $\to$ L: $\times 3.785$	L $\to$ US gal: $\div 3.785$	US fl oz $\to$ mL: $\times 29.57$	mL $\to$ US fl oz: $\div 29.57$
British gal $\to$ L: $\times 4.546$	L $\to$ British gal: $\div 4.546$		

Estimate first. A US gallon is a bit less than 4 L. A British gallon is about 4.5 L. A US fluid ounce is about 30 mL. If your answer is far from the estimate, you probably multiplied when you should have divided.

Example 6

Estimate, then convert. Round to the nearest hundredth (or tenth for mL).

	Estimate	**Work**	**Answer**
(a) 10 US gal to litres			
(b) 60 L to US gallons			
(c) a 12 US fl oz can to mL			
(d) a 500 mL bottle to US fl oz			

(e) Look at a can of pop and a 500 mL water bottle from a store. What do the labels say, and do they match (c) and (d)?

Example 7

Sana drives her car from Calgary to Montana. Her gas tank holds 52 L. In Montana, gas costs \$3.29 USD per US gallon.

(a) How many US gallons does her tank hold? Round to the nearest hundredth.

(b) How much will it cost to fill the tank from empty, in US dollars?

(c) Find the price of gas per litre in US dollars. Round to three decimals.

(d) Use 1 USD = 1.38 CAD. Is gas in Montana cheaper than \$1.45 CAD per litre in Calgary? By how much per litre?

Kitchen measures. Canadian measuring cups and spoons are metric. American recipes use US cups. The two cups are close but not the same.

Canadian (metric) kitchen: 1 teaspoon (tsp) = 5 mL 1 tablespoon (tbsp) = 15 mL = 3 tsp 1 cup = 250 mL
US recipes: 1 US cup = 8 US fl oz $\approx$ 237 mL

Example 8

Marco finds a pancake recipe on an American website. It needs $1\frac{1}{2}$ US cups of milk, $\frac{3}{4}$ US cup of oil and 2 tsp of vanilla. Use 1 US cup $\approx$ 237 mL.

(a) Convert each amount to millilitres. Round to the nearest millilitre.

Ingredient	**Work**	**mL**
$1\frac{1}{2}$ US cups of milk		
$\frac{3}{4}$ US cup of oil		
2 tsp of vanilla		

(b) Marco measures the milk with his Canadian 250 mL cup instead: $1\frac{1}{2}$ metric cups. How much extra milk is that?

(c) Finish the sentence.

For pancakes, using metric cups is _____ because _____

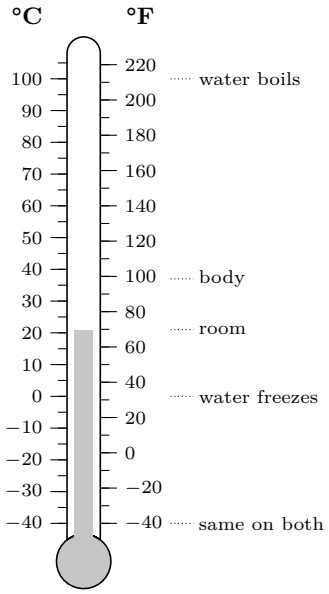
Lesson 3: Temperature: Celsius and Fahrenheit

Reading the two temperature scales, converting both ways with a formula, estimating in your head, and comparing temperatures and temperature changes.

1. Two Temperature Scales

Temperature tells how hot or cold something is. Canada uses **degrees Celsius** (°C), the SI scale. The United States uses **degrees Fahrenheit** (°F), the imperial scale. You see Fahrenheit on US weather reports, US recipes and ovens, and on some tools and manuals made in the US.

A **double thermometer** shows both scales side by side. Each height on it is *one* temperature with two names.



Referents (temperatures to remember)		
	°C	**°F**
Water boils	100	212
Hot oven (baking)	180	350
Normal body temperature	37	98.6
Room temperature	20 to 22	68 to 72
Water freezes	0	32
Very cold winter day	−40	−40

From freezing to boiling is **100 Celsius degrees** but **180 Fahrenheit degrees**. So a Celsius degree is bigger: **a change of 1 Celsius degree is a change of 1.8 Fahrenheit degrees.** The scales also start in different places: water freezes at 0 on one and at 32 on the other.

Example 1

(a) Use the double thermometer to estimate each temperature on the other scale.

(i) 10 °C ≈ _____

(ii) 80 °F ≈ _____

(iii) −20 °C ≈ _____

(b) Each temperature is missing its unit. Write °C or °F so that it makes sense.

(i) A mug of hot chocolate: 60 _____

(ii) A sunny spring day in Calgary: 15 _____

(iii) A warm swimming pool: 80 _____

(iv) A child with a fever: 39 _____

2. Celsius to Fahrenheit

To change °C to °F, multiply by 1.8 (Fahrenheit degrees are smaller, so there are more of them), then add 32 (because water freezes at 32 °F, not 0).

Celsius to Fahrenheit: $F = 1.8 \times C + 32$ (the same as $F = \frac{9}{5}C + 32$, because $\frac{9}{5} = 1.8$)
Step 1: Multiply the °C temperature by 1.8. **Step 2:** Add 32.

Calculator. 1.8 [×] 25 [+] 32 [ENTER] shows 77, so 25 °C = 77 °F. For a temperature below zero, use the negative key [(−)], not the subtract key [−]: 1.8 [×] [(−)] 15 [+] 32 [ENTER] shows 5.

Example 2

Change each temperature to degrees Fahrenheit.

	°C	**Work** $(1.8 \times C + 32)$	**°F**
(a) A hot summer afternoon	25 °C		
(b) Normal body temperature	37 °C		
(c) A cold January morning	−15 °C		
(d) A pizza oven	230 °C		
(e) A warm bedroom	22.5 °C		
(f) A bitter cold night	−30 °C		

Example 3

Amira's family is staying at a hotel in Montana. The thermostat in the room is in °F.

(a) At home, Amira keeps her room at 21 °C. What should she set the hotel thermostat to? Round to the nearest degree.

(b) Her phone says it is −8 °C at home in Calgary. Her cousin in Montana asks what that is in °F. What should Amira tell her?

3. Fahrenheit to Celsius

To go the other way, **undo** the steps in reverse order: first subtract 32, then divide by 1.8.

Fahrenheit to Celsius: $C = (F - 32) \div 1.8$ (the same as $C = \frac{5}{9}(F - 32)$)
Step 1: Subtract 32 from the °F temperature. **Step 2:** Divide by 1.8.

Calculator. Use brackets so the calculator subtracts first: [(] 98.6 [−] 32 [)] [÷] 1.8 [ENTER] shows 37.

Watch out. Without the brackets, 98.6 [−] 32 [÷] 1.8 [ENTER] divides first (order of operations) and shows 80.82..., which is wrong. Unless a question says otherwise, round temperatures to the nearest **tenth** of a degree.

Example 4

Change each temperature to degrees Celsius. Round to the nearest tenth of a degree.

	°F	**Work** $((F - 32) \div 1.8)$	**°C**
(a) A beach day in California	86 °F		
(b) A very hot day	100 °F		
(c) A deep freezer	0 °F		
(d) A winter night in Minnesota	−13 °F		
(e) An oven for roasting	425 °F		
(f) A cool fall morning	59 °F		

Example 5

Lucas is baking his aunt's cookie recipe from Arizona. It says "Bake at 375 °F". His oven is in °C and can be set in steps of 5 degrees.

(a) Change 375 °F to °C, to the nearest tenth. What should Lucas set his oven to?

(b) A US meat thermometer says chicken is safe when it reaches 165 °F inside. What is that in °C, to the nearest degree?

4. Estimating in Your Head

You will not always have a calculator, for example when a friend tells you a US weather forecast. These quick rules round 1.8 up to 2 and 32 down to 30:

Quick estimates:	$F \approx 2 \times C + 30$ (double, then add 30)	$C \approx (F - 30) \div 2$ (subtract 30, then halve)

The estimate is very close for ordinary weather. It gets worse for very hot or very cold temperatures. For cooking, medicine or a job, use the exact formula.

Example 6

(a) Complete the table.

°C	**Estimate** $2C + 30$	**Exact** $1.8C + 32$	**How far off?**
−20			
0			
10			
20			
30			
200			

(b) Finish the sentences.

The estimate is exact at _____.

It is within a few degrees for _____ temperatures,

but it is far off for _____ temperatures, like an oven.

(c) In Seattle, the forecast says 54 °F. Estimate the temperature in °C. Then find the exact value. Is the estimate good enough to decide whether to wear a jacket?

(d) A US recipe says to roast vegetables at 450 °F. Estimate in °C, then find the exact value. Is the estimate good enough?

5. Comparing Temperatures and Temperature Changes

To **compare** two temperatures, change them to the **same scale** first.

A **temperature change** is how much the temperature goes up or down. Subtract: higher − lower. With a negative temperature, count up to 0 and then past it: from -21 up to 7 is $21 + 7 = 28$ degrees.

To change a *temperature change* to the other scale, only multiply or divide by 1.8. Do **not** add or subtract 32: the 32 is about where the scales start, and a change does not depend on where they start.

Change in °F = 1.8× change in °C	**Change in °C** = change in °F ÷ 1.8
−40 is the same on both scales: $1.8 \times (-40) + 32 = -72 + 32 = -40$.	

Example 7

(a) On the same morning it is −25 °C in Edmonton and −15 °F in Minneapolis. Which city is colder, and by how much?

(b) Food safety rules say a fridge should be at 4 °C or colder. A US-made fridge thermometer reads 38 °F. Is the fridge cold enough?

Example 8

A **chinook** is a warm, dry wind that blows down from the Rocky Mountains in winter. One morning in Calgary it is −21 °C. A chinook arrives, and by the afternoon it is 7 °C.

(a) By how many Celsius degrees did the temperature rise?

(b) By how many Fahrenheit degrees did it rise?

(c) Check (b): change both temperatures to °F and subtract.

Watch out. If you add 32 to the change, you get $50.4 + 32 = 82.4$, which is far too big. Only actual temperatures (readings on a thermometer) use the +32 or −32.

Lesson 4: Mass: Grams, Kilograms, Ounces, Pounds and Tons

Choosing a sensible unit of mass, converting within the metric and imperial systems, working with pounds and ounces, and reading scales.

1. Mass, Weight and Referents

Mass is how much matter (stuff) is in an object. In everyday speech we say **weight**: "weigh the apples", "my weight". In this booklet, *mass* and *weight* mean the same thing. (A scientist would say weight is the pull of gravity. On the Moon you would weigh less, but your mass would not change.)

Canada uses **SI** (metric) units of mass. You will still see **imperial** units on meat counters, at the gym, on baby scales and in trucking.

SI units of mass		
milligram	mg	medicine and vitamins (a pill might have 500 mg)
gram	g	a paper clip is about 1 g; a toonie is about 7 g
kilogram	kg	1 L of water is about 1 kg; a bag of sugar is often 2 kg
tonne	t	1000 L of water is about 1 t; a small car is about 1.3 t
Imperial units of mass		
ounce	oz	a slice of bread is about 1 oz; a hockey puck is about 6 oz
pound	lb	a block of butter is 1 lb
ton	(tn)	a big bison bull is about 1 ton

Watch out. A **tonne** (t) is metric: 1 t = 1000 kg. A **ton** is imperial: 1 ton = 2000 lb. They are different units, even though they sound the same. (The imperial ton is also called a **short ton**.)

Example 1

(a) Choose a sensible unit for the mass of each object.

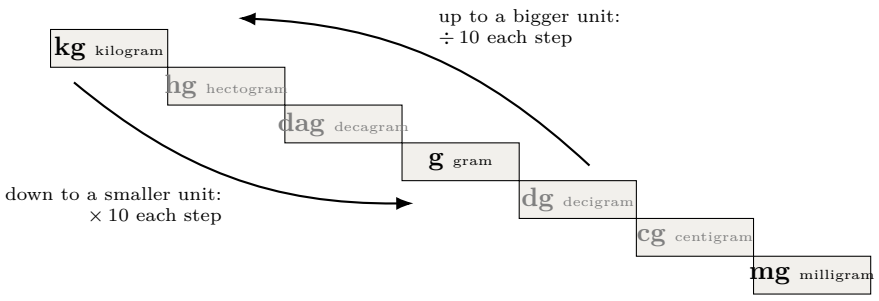
Object	**SI unit**	**Imperial unit**
a bag of sugar		
a vitamin pill		
a school bus		
a strawberry		
a Grade 10 student		

(b) Circle the most sensible mass.

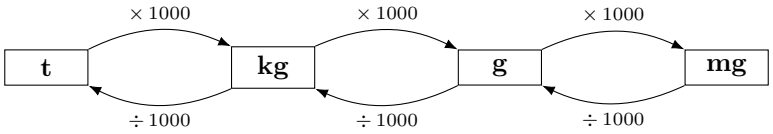
(i) a large pumpkin:	6 g	6 kg	6 t
(ii) a smartphone:	190 mg	190 g	190 kg
(iii) a pickup truck:	2.5 kg	2.5 t	250 t
(iv) a big apple:	8 oz	8 lb	8 tons

2. Converting SI Units of Mass

SI units use the same **prefixes** as length: **kilo** means 1000, **milli** means $\frac{1}{1000}$. Each step on the staircase is 10 times the next one down.



The grey units are rarely used for mass. The ones you need are 1000 times apart, so each jump is three steps:



1 t = 1000 kg
1 kg = 1000 g
1 g = 1000 mg

Bigger unit → smaller unit: multiply. **Smaller unit → bigger unit: divide.**

Check. When you change to a **smaller** unit you need **more** of them, so the number gets bigger: 2.4 kg = 2400 g. Multiplying or dividing by 1000 moves the decimal point **3 places**.

Example 2

Convert. Show whether you multiply or divide.

Change	Work	Answer
(a) 2.4 kg to grams		
(b) 750 g to kilograms		
(c) 3500 kg to tonnes		
(d) 0.6 t to kilograms		
(e) 250 mg to grams		
(f) 45 g to kilograms		
(g) 1.2 g to milligrams		
(h) 2.5 kg to milligrams (two jumps)		

Example 3

Ruby is baking bread for a bake sale. Each loaf uses 650 g of flour. She bakes 6 loaves. Flour comes in 2.5 kg bags.

(a) How much flour does she need, in kilograms?

(b) How many bags of flour must she buy?

(c) How much flour will be left over? Is it enough for one more loaf?

Example 4

A flatbed truck has a mass of 4.8 t when it is empty. It is loaded with 14 pallets of patio stones. Each pallet has a mass of 925 kg. A sign on a bridge says the limit is 18 t.

(a) Find the mass of the pallets, in tonnes.

(b) Can the loaded truck cross the bridge?

(c) Could the driver add one more pallet? Explain.

Watch out. Change to the **same unit** before you add or compare. Adding 4.8 t and 12 950 kg as "4.8 + 12 950" makes no sense.

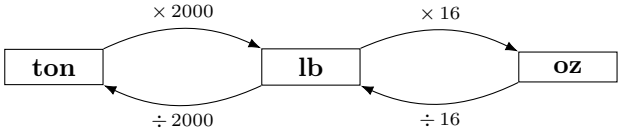
Example 5

(a) Sliced turkey costs \$2.38 per 100 g at the deli. Find the price per kilogram, and the cost of 350 g.

(b) Each vitamin C tablet has 500 mg of vitamin C. A health guide says an adult should take no more than 2 g a day. What is the most tablets a day?

3. Converting Imperial Units of Mass

Imperial units are **not** based on 10. You have to remember the numbers:



1 ton = 2000 pounds (lb)
1 pound (lb) = 16 ounces (oz)

Bigger unit → smaller unit: multiply. Smaller unit → bigger unit: divide.

Masses are often written in **pounds and ounces**, like 3 lb 7 oz. To change it to ounces, change the pounds to ounces and add the extra ounces: $3 \times 16 + 7 = 55$ oz.

Ounces to pounds and ounces. Step 1: divide by 16; the whole number is the pounds. Step 2: the ounces left over are the decimal part $\times 16$ (or subtract the ounces in those pounds).

Calculator. 70 oz: 70 [÷] 16 [ENTER] shows 4.375, so 4 lb. Then [−] 4 [ENTER] shows 0.375, and [×] 16 [ENTER] shows 6. So 70 oz = 4 lb 6 oz. (Check: $4 \times 16 + 6 = 70$.)

Watch out. The decimal part of a pound is **not** ounces. 1.5 lb is *not* 1 lb 5 oz. Half a pound is 8 oz, so 1.5 lb = 1 lb 8 oz.

Example 6

Convert.

Change	Work	Answer
(a) 5 lb to ounces		
(b) 3 lb 7 oz to ounces		
(c) 72 oz to pounds		
(d) 70 oz to pounds and ounces		
(e) 3.25 lb to pounds and ounces		
(f) 10 oz to pounds		
(g) 3.5 tons to pounds		
(h) 9000 lb to tons		

Example 7

(a) A landscaper has 2.75 tons of gravel to move into a backyard. Her wheelbarrow holds about 125 lb. How many loads will it take?

(b) A rancher feeds each of her 45 cows about 28 lb of hay a day. How many tons of hay does she need for 30 days? Her round bales are 1400 lb each. How many bales is that?

4. Working with Pounds and Ounces

To add or subtract masses in pounds and ounces, line up **pounds under pounds** and **ounces under ounces**, like adding hours and minutes.

Adding: add the ounces, add the pounds. If you have 16 oz or more, trade 16 oz for 1 lb.
Subtracting: if there are not enough ounces on top, borrow 1 lb and turn it into 16 oz.
Multiplying: multiply the pounds and the ounces, then trade every 16 oz for 1 lb.

Example 8

(a) 5 lb 12 oz + 3 lb 9 oz

(b) 7 lb 3 oz − 2 lb 10 oz

(c) $4 \times$ (1 lb 11 oz)

Example 9

(a) Owen is packing for an overnight hike. His tent is 4 lb 9 oz, his sleeping bag is 2 lb 14 oz, his stove and pot are 1 lb 3 oz, and his food is 3 lb 11 oz. He wants the total under 12 lb. Is it? By how much is he over or under?

(b) A puppy named Bear weighed 6 lb 13 oz at 8 weeks old. At 12 weeks he weighed 11 lb 4 oz. How much did he gain?

Prices by the pound or ounce. To compare sizes, change both to the **same unit** (usually ounces), then find the unit price, just like in Unit 1. Sometimes part of what you buy is thrown away (peel, bones, fat). The **true cost** per pound is what you paid ÷ the pounds you can actually use.

True cost per pound = total price paid ÷ usable pounds (change pounds and ounces to decimal pounds first: 9 oz = 9 ÷ 16 = 0.5625 lb)

Example 10

(a) A 12 oz bag of coffee costs $10.99. A 2 lb bag of the same coffee costs $26.99. Which is the better buy?

(b) Nadia buys a watermelon that weighs 9 lb 4 oz, at $0.80 per lb. After she cuts off the rind, she has 5 lb 9 oz of fruit. What did she pay, and what is her true cost per pound of fruit?

5. Reading Scales

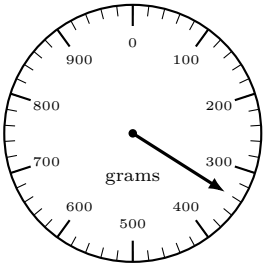
A **scale** measures mass. To read a scale with a dial or a pointer:

- Step 1: find two numbers next to each other and their difference.
- Step 2: count the **spaces** (not the lines) between them. One space is worth difference ÷ spaces.
- Step 3: start at the last number before the pointer and count on.

A **digital scale** shows the number for you. Its **TARE** (or ZERO) button sets the scale to 0 with a bowl on it, so you weigh only what you put in the bowl. Without it, subtract: mass of food = total − mass of the bowl.

Example 11

Read each scale.



(a) Kitchen scale. What is one small space worth? _____

What does it read? _____

(b) Bathroom scale (pounds). One small space: _____ Reading: _____



(c) Produce scale (pounds). One small space: _____ Reading: _____



(d) Jae puts a mixing bowl full of flour on a digital scale. It shows 1.24 kg. The empty bowl has a mass of 315 g. How much flour is in the bowl?

Lesson 5: Converting Mass Between SI and Imperial

Changing pounds and ounces to kilograms and grams (and back), estimating in your head, and using it for baggage, shipping, loads and meat prices.

1. The Conversion Facts and Quick Estimates

In Canada, most things are labelled in grams and kilograms, but many people still use **pounds** (lb) and **ounces** (oz): bathroom scales, gym plates, newborn babies, US recipes and some flyers. To change between the two systems, you need a few **conversion facts**. They are **approximate** (the symbol ≈ means "is about").

Conversion facts	**Inside each system** (exact)	**Referents**
1 lb ≈ 0.454 kg (= 454 g)	1 lb = 16 oz	a 1 lb block of butter is 454 g
1 kg ≈ 2.2 lb	1 ton = 2000 lb	a 1 kg bag of sugar is a bit more than 2 lb
1 oz ≈ 28.35 g	1 kg = 1000 g	a slice of bread is about 1 oz (about 30 g)
1 ton ≈ 0.907 t	1 tonne (t) = 1000 kg	a ton is a bit less than a tonne

In everyday speech people say **weight** when they mean mass ("my bag weighs 20 kg"). In this lesson the two words mean the same thing. A **ton** here is the short ton (2000 lb). Do not mix it up with a **tonne** (t), which is 1000 kg.

Quick estimates in your head. A kilogram is a bit more than 2 pounds. A pound is a bit less than half a kilogram.

kg → lb: double it, then add 10% of that. e.g. 20 kg: $20 \times 2 = 40$, $40 + 4 = 44$ lb
lb → kg: halve it, then take away 10% of that. e.g. 120 lb: $120 \div 2 = 60$, $60 - 6 = 54$ kg
oz → g: about 30 g for each ounce. e.g. 5 oz $\approx 5 \times 30 = 150$ g

Example 1

(a) Circle the heavier mass in each pair. Use the conversion facts.

Pair	Heavier	How I know
(i) 1 kg or 1 lb		
(ii) 4 lb or 2 kg		
(iii) 1 oz or 25 g		
(iv) 1 ton or 1 tonne		
(v) 8 oz or 250 g		

(b) Estimate in your head. Show your thinking.

(i) A suitcase has a mass of 30 kg. About how many pounds is it?

(ii) Jacob weighs 160 lb. About how many kilograms is that?

(iii) A steak is 6 oz. About how many grams is it?

2. Converting with a Unit Factor

For an exact answer, multiply by a **unit factor**: a fraction made from a conversion fact, like $\frac{0.454 \text{ kg}}{1 \text{ lb}}$. Its top and bottom are the same mass, so it equals 1. Multiplying by it changes the units but not the mass.

Step 1. Write the amount you have, with its unit.
Step 2. Multiply by a unit factor with the unit you **have** on the **bottom** and the unit you **want** on the **top**.
Step 3. Cross out the unit that appears on the top and the bottom. Calculate, then round.
Step 4. Check with a quick estimate.

For example, 25 lb to kilograms: $25 \cancel{\text{lb}} \times \frac{0.454 \text{ kg}}{1 \cancel{\text{lb}}} = 25 \times 0.454 = 11.35$ kg. Check: half of 25 is 12.5, take away 1.25: about 11.3 kg.

And 500 g to ounces: $500 \cancel{\text{g}} \times \frac{1 \text{ oz}}{28.35 \cancel{\text{g}}} = 500 \div 28.35 \approx 17.6$ oz. (The fact is on the bottom, so you **divide**.)

You can also use a proportion, as in Unit 1: $\frac{1 \text{ lb}}{0.454 \text{ kg}} = \frac{25 \text{ lb}}{x}$ gives $x = 25 \times 0.454 = 11.35$ kg.

Watch out. Ask: should the number get bigger or smaller? A kilogram is **bigger** than a pound, so you need **more** pounds than kilograms: 10 kg is about 22 lb, not about 4.5 lb. In the same way, a gram is tiny, so a mass in grams is a much bigger number than the same mass in ounces.

Calculator. 25 [×] 0.454 [ENTER] shows 11.35. 500 [÷] 28.35 [ENTER] shows 17.63668…, about 17.6.

Example 2

Convert. Round to the nearest tenth.

Convert	Work (unit factor)	Answer
(a) 25 lb to kg		
(b) 68 kg to lb		
(c) 12 oz to g		
(d) 500 g to oz		
(e) 3 tons to tonnes		
(f) 4.5 t to tons		

Pounds and ounces. A mass like 7 lb 6 oz has two units. Change it to **one** unit first: either all pounds (ounces ÷ 16) or all ounces (pounds × 16, then add the ounces).

Watch out. 7 lb 6 oz is **not** 7.6 lb. There are 16 oz in a pound, not 10. 6 oz $= 6 \div 16 = 0.375$ lb, so 7 lb 6 oz $= 7.375$ lb.

Example 3

(a) Baby Amara was born at 7 lb 6 oz. Find her mass in kilograms, to the nearest hundredth, two ways.

(i) Change to pounds, then to kilograms.

(ii) Change to ounces, then to grams, then to kilograms.

(b) A bag of dog food is 9 kg. Find its mass in pounds and ounces, to the nearest ounce.

3. Limits: Baggage, Shipping and Loads

Airlines, couriers, trucks and bridges all have **mass limits**. Often the limit is in one system and your scale is in the other. Convert so both masses are in the **same unit**, then compare.

Round down for "how many". If the question asks how many whole items fit under a limit, **round down**. 24.8 bags means only 24 bags fit: the 25th bag would go over.

Example 4

An airline lets each passenger check one bag of up to 23 kg. Tamika's bathroom scale shows her suitcase at 53 lb.

(a) Estimate: is her bag over the limit?

(b) Find the bag's mass in kilograms, to the nearest tenth. By how much is it over?

(c) Her carry-on bag may be up to 7 kg. It now weighs 11 lb. Can she move 1.5 kg of things from the suitcase into the carry-on and still be within both limits?

Example 5

Jun ships candles from her online shop. Each candle has a mass of 14 oz. Her courier's prices are below.

Mass of parcel	**Price**
up to 1 kg	$13.95
over 1 kg, up to 2.5 kg	$18.40
over 2.5 kg, up to 5 kg	$24.75

(a) A parcel holds 2 candles, a 250 g gift box of soap, and 180 g of box and packing. Find its mass in kilograms. What is the price to ship it?

(b) With the same soap and packing, what is the greatest number of candles Jun can put in one parcel and still pay $18.40?

Example 6

A rental dump trailer can carry at most 3 t (tonnes). Gravel has a mass of about 1.4 tons per cubic yard.

(a) Can the trailer safely carry 2 cubic yards of gravel?

(b) Marco needs 5 cubic yards for his driveway. The supplier loads whole cubic yards only. How many trips will he need?

4. Buying by the Pound or by the Kilogram

Meat, fish, cheese and produce are often priced **per pound** in flyers (\$/lb) and **per kilogram** at the till (\$/kg). To compare, change both to the same unit price.

A kilogram is about 2.2 pounds, so **1 kg costs about 2.2 times as much as 1 lb**. The price per kg is always the **bigger** number.

price per kg = price per lb × 2.2	**price per lb** = price per kg ÷ 2.2

Watch out. Converting a *price* goes the opposite way to converting a *mass*. To change 5 lb to kg you multiply by 0.454, but to change \$5.00/lb to \$/kg you multiply by 2.2. Think: "a kilogram is more meat, so it costs more." Stores use a more exact factor, so their numbers may be a cent or two different from yours.

Example 7

(a) Complete the table. Round to the nearest cent.

Item	Price per lb	Work	Price per kg
(i) Ground beef	\$5.49		
(ii) Pork tenderloin			\$14.30
(iii) Sweet potatoes			\$2.53

(b) Store A sells strip loin steak for \$9.99/lb. Store B sells the same steak for \$21.50/kg. Which store has the better price, and by how much per kilogram?

Example 8

The volleyball team is having a barbecue. They need 30 quarter-pound burger patties (each patty is $\frac{1}{4}$ lb of ground beef). Ground beef costs \$12.00 per kg.

(a) How many pounds of beef do they need?

(b) How many kilograms is that?

(c) How much will the beef cost?

(d) The store only sells the beef in 1 kg packages. How many packages must they buy, and what will they pay?

Lesson 6: From Volume to Mass: Bushels, Tonnes and Conversion Factors

Using a conversion factor to change an amount of space (litres, bushels, cubic feet) into a mass (kilograms, tonnes), and back, for liquids, gravel and Prairie grain.

1. Volume and Mass Are Different

Volume is how much space something takes up (L, m^3, ft^3). **Mass** is how heavy it is (kg, t, lb). A bucket of sand and the same bucket of popcorn have the same volume, but the sand is much heavier. So to change a volume into a mass, you need to know **what the stuff is**.

A **conversion factor** tells you how much mass is in one unit of volume of one material, for example "milk: 1.03 kg per L". Water is the easy one: the metric system was built so that 1 mL of water has a mass of about 1 g.

Water:	1 mL $\approx$ 1 g 1 L $\approx$ 1 kg 1 m^3 = 1000 L $\approx$ 1000 kg = 1 t
Any material:	mass = volume $\times$ (mass per unit of volume) volume = mass $\div$ (mass per unit of volume)

Watch out. The units must match. If the factor is "kg per L", the volume must be in litres (change mL to L first), and the answer is in kg.

Example 1

Find each mass. Use the conversion factor given.

Material	Factor	Volume	Work	Mass
(a) Water in a jug	1 kg per L	4 L		
(b) Milk	1.03 kg per L	4 L		
(c) Gasoline in a full tank	0.74 kg per L	60 L		
(d) Honey in a pail	1.42 kg per L	2.5 L		
(e) Water in a bottle	1 g per mL	355 mL		

(f) A 60 L tank of gasoline weighs less than 60 L of water. How do you know from the factors?

Round down or round up?. If a load **must not go over** a limit, round the amount **down**, so you stay under the limit. If you need **enough** loads, trips or bags to do the whole job, round **up**.

Example 2

Ravi is building a gravel driveway in Cochrane. Gravel has a mass of about 1.7 t per m^3. The driveway needs 8 m^3 of gravel. The landscaper's dump truck can carry at most 6 t.

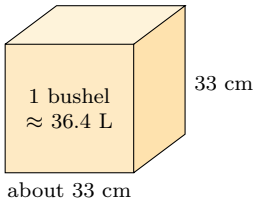
(a) What is the mass of the gravel Ravi needs?

(b) How many cubic metres can the truck carry in one load? Round down to the nearest tenth.

(c) How many loads are needed?

2. What Is a Bushel?

A **bushel** (bu) is an old imperial unit of **volume** for grain and other dry crops. One bushel is 8 imperial (British) gallons. That is about the space inside a box 33 cm on each side.



1 bushel ≈ 36.4 L ≈ 1.28 ft^3 ≈ 2220 in^3 1 m^3 ≈ 27.5 bu

Prairie farmers still talk in bushels ("this bin holds 3000 bushels"). But grain elevators **weigh** grain and pay by the **tonne**. So farmers must change bushels to tonnes.

A bushel of heavy grain weighs more than a bushel of light grain. So each crop has its own **conversion factor** in **bushels per tonne** (bu/t): the number of bushels it takes to make **1 tonne** (1000 kg).

Crop	Wheat	Canola	Barley	Oats
Bushels per tonne	36.7	44.1	45.9	64.8

Factors in this booklet are rounded examples. Grain buyers use exact tables for each crop.

A **bigger** factor means a **lighter** bushel: you need more bushels to make a tonne.

Mass of one bushel: kg per bushel $= 1000 \div$ factor e.g. wheat: $1000 \div 36.7 \approx 27.2$ kg

US or British?. A US bushel is a little smaller (about 35.2 L), just like a US gallon is smaller than a British gallon. In this booklet a bushel is the Canadian (imperial) bushel.

Example 3

(a) Find the mass of one bushel of each crop. Round to the nearest tenth of a kilogram.

Crop	**bu/t**	**Work**	**kg/bu**
Wheat	36.7	$1000 \div 36.7$	27.2
Canola	44.1		
Barley	45.9		
Oats	64.8		

(b) Which crop is the heaviest per bushel? Which is the lightest?

Heaviest _____ Lightest _____

(c) Two identical trucks are filled to the same level, one with wheat and one with oats. Which truck is heavier? Explain.

3. Converting Between Bushels and Tonnes

The factor says how many bushels make 1 tonne. So:

Tonnes → bushels: multiply by the factor. $12 \text{ t} \times \dfrac{36.7 \text{ bu}}{1 \text{ t}} = 440.4 \text{ bu}$

Bushels → tonnes: divide by the factor. $850 \text{ bu} \times \dfrac{1 \text{ t}}{44.1 \text{ bu}} = 850 \div 44.1 \approx 19.27 \text{ t}$

Write the factor as a fraction so the unit you start with **cancels** (t on the bottom to cancel t; bu on the bottom to cancel bu).

Check. The number of bushels is always **bigger** than the number of tonnes (about 35 to 65 times bigger). If your tonnes come out bigger than your bushels, you went the wrong way.

Rounding. Unless a question says otherwise, round bushels to the nearest whole bushel and tonnes to two decimals (the nearest 10 kg).

Example 4

Convert. Use the factors in the table above.

Crop	Factor	Change	Work	Answer
(a) Wheat	36.7 bu/t	12 t to bu		
(b) Canola	44.1 bu/t	850 bu to t		
(c) Barley	45.9 bu/t	2000 bu to t		
(d) Oats	64.8 bu/t	6.5 t to bu		

Example 5

(a) A hobby farm near Olds buys 150 bushels of barley for its cattle. Find the mass in tonnes, in kilograms (to the nearest kg) and in pounds. Use 1 kg $\approx$ 2.2 lb.

(b) Nadia's grain truck may legally carry 14.5 t. Can she haul 560 bushels of wheat in one load? What is the most wheat (whole bushels) she can haul?

4. Price per Bushel and Price per Tonne

Elevators post grain prices in **dollars per tonne**, but many farmers still think in **dollars per bushel**. One tonne of wheat is 36.7 bushels, so the price of one tonne is shared by 36.7 bushels.

\$ per bushel = \$ per tonne ÷ factor **\$ per tonne** = \$ per bushel × factor

(The price per tonne is always the bigger number, because a tonne is many bushels.)

Money. Keep all the decimals in your calculator and round to the nearest cent only at the end.

Example 6

Change each price. Round to the nearest cent.

Crop	Factor	Price	Work	New price
(a) Canola	44.1 bu/t	\$680.00/t		
(b) Wheat	36.7 bu/t	\$7.80/bu		
(c) Oats	64.8 bu/t	\$4.25/bu		
(d) Barley	45.9 bu/t	\$255.00/t		

Example 7

Liam is selling barley (45.9 bu/t). Buyer A offers \$5.70 per bushel. Buyer B offers \$260.00 per tonne.

(a) Change Buyer A's price to dollars per tonne. Which buyer pays more?

(b) Liam sells 3000 bushels. How much more money does he get from the better buyer?

5. Grain Bins: From Volume to Bushels

A round steel grain bin is a **cylinder**, so the grain inside has volume $V = \pi r^2 h$ (Lesson 1), where h is how deep the grain is. To find what the bin holds, change the volume to bushels, then to tonnes.

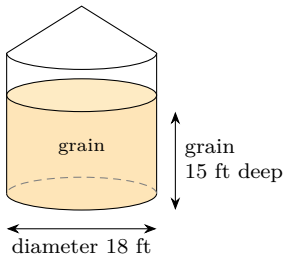
volume → bushels → tonnes

$\text{ft}^3 \to$ bu: divide by 1.28 (1 bu ≈ 1.28 ft^3) $\text{m}^3 \to$ bu: multiply by 27.5 (1 $\text{m}^3 \approx 27.5$ bu)

Check. The number of bushels is a bit **less** than the number of cubic feet (about $\frac{4}{5}$ of it), but about 27 times **more** than the number of cubic metres. Real bins have a cone-shaped pile on top. In this booklet the grain is level.

Example 8

A wheat bin on a farm near Lethbridge has a diameter of 18 ft. The wheat is level and 15 ft deep.



(a) Find the volume of the wheat, to the nearest cubic foot.

(b) How many bushels is that?

(c) Find the mass of the wheat (36.7 bu/t), to the nearest tenth of a tonne.

(d) A truck hauls 600 bushels per trip. How many trips will it take to empty the bin?

Example 9

A grain truck box is 6.1 m long and 2.4 m wide. It is filled level to a depth of 1.5 m. The truck may carry at most 14.5 t.

(a) Find the volume of grain in the box.

(b) How many bushels is that, to the nearest bushel?

(c) If the box is full of canola (44.1 bu/t), what is the mass? Is the load legal?

(d) If the box is full of wheat (36.7 bu/t), is the load legal? What should the driver do?

Lesson 7: Unit 4 Review

Every rule from the unit on one page, worked examples that mix the topics, and a practice unit test.

1. The Unit on One Page

Use this page to study. Each row is one skill: what to do, and a short example. The conversion numbers marked ≈ are rounded, so answers that use them are close, not exact.

Skill	What to do	Example
Volume of a prism	V = (area of base) $\times h$. A box: $V = lwh$. A triangular prism: base area $= \frac{1}{2}bh$ of the triangle. Use the same unit for every length. Volume is in **cubic units** (cm^3, m^3, ft^3).	A box 2 m $\times$ 1.5 m $\times$ 0.4 m: $V = 2 \times 1.5 \times 0.4 = 1.2$ m^3
Volume of a cylinder	$V = \pi r^2 h$. The radius is half the diameter: $r = d \div 2$.	$r = 10$ cm, $h = 30$ cm: $V = \pi \times 10^2 \times 30 \approx 9425$ cm^3
Cubic units	1 m^3 = 1 000 000 cm^3; 1 ft^3 = 1728 in^3 ($12 \times 12 \times 12$); 1 yd^3 = 27 ft^3 ($3 \times 3 \times 3$).	54 $ft^3 \div 27 = 2$ yd^3
Capacity (SI)	1 mL = 1 cm^3; 1 L = 1000 mL = 1000 cm^3; 1 m^3 = 1000 L.	9425 cm^3 = 9425 mL = 9.425 L
Capacity (imperial)	US: 1 gal = 4 qt = 8 pt = 16 cups = 128 fl oz; 1 US gal $\approx$ 3.785 L; 1 US fl oz $\approx$ 29.57 mL. British: 1 gal $\approx$ 4.546 L (about 20% bigger); 1 pint = 20 fl oz $\approx$ 568 mL. Metric cup = 250 mL; US cup $\approx$ 237 mL.	5 US gal $\approx 5 \times 3.785 = 18.925$ L
Temperature	$F = 1.8 \times C + 32$; $C = (F - 32) \div 1.8$. Quick estimate: $F \approx 2 \times C + 30$. Water freezes at 0 °C (32 °F) and boils at 100 °C (212 °F); body 37 °C.	25 °C: $1.8 \times 25 + 32 = 77$ °F
Mass (SI)	1 g = 1000 mg; 1 kg = 1000 g; 1 t = 1000 kg. To a smaller unit, multiply; to a bigger unit, divide.	2.75 kg = 2750 g
Mass (imperial)	1 lb = 16 oz; 1 ton = 2000 lb. When you add, change every 16 oz into 1 lb. When you subtract, borrow 1 lb as 16 oz.	4 lb 10 oz + 2 lb 11 oz = 6 lb 21 oz = 7 lb 5 oz
SI $\leftrightarrow$ imperial mass	1 lb $\approx$ 0.454 kg; 1 kg $\approx$ 2.2 lb; 1 oz $\approx$ 28.35 g; 1 ton $\approx$ 0.907 t. Check: the number of pounds is always **bigger** than the number of kilograms.	50 lb $\approx 50 \times 0.454 = 22.7$ kg
Bushels and tonnes	The factor (bushels per tonne, bu/t) is given for each crop. tonnes $\times$ factor = bushels; bushels $\div$ factor = tonnes.	2205 bu of canola at 44.1 bu/t: $2205 \div 44.1 = 50$ t

Remember. Change every length to the same unit **before** you find a volume (4 in $= \frac{4}{12}$ ft). Use the radius, not the diameter, in $\pi r^2 h$. Say which gallon or cup you mean: US and British are different. Estimate first, so you can spot a wrong calculator answer. Round only at the **end**, and round **up** when you buy whole bags, cans or loads.

2. Volume and Capacity

Example 1

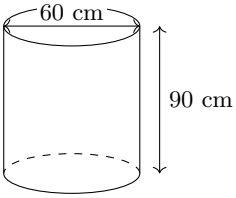
A raised garden bed is a box 2.4 m long, 1.2 m wide and 0.35 m deep.

(a) How many cubic metres of soil fill the bed?

(b) Garden soil is sold in 25 L bags. How many bags are needed?

Example 2

A rain barrel is a cylinder 60 cm across (diameter) and 90 cm tall.



(a) Find the volume of the barrel in cubic centimetres.

(b) How many litres does it hold?

(c) An American store lists the barrel in US gallons. About how many US gallons is it? (1 US gal $\approx$ 3.785 L)

(d) A British store lists it in British gallons (1 gal $\approx$ 4.546 L). Find that number. Why is it smaller than (c)?

Example 3

Dev is pouring a concrete pad for a shed. The pad is 10 ft by 12 ft and 4 in thick.

(a) Find the volume of the pad in cubic feet.

(b) Change the volume to cubic yards.

(c) The concrete plant sells in half cubic yards (0.5, 1, 1.5, . . .) at $210.00 per cubic yard. How much should Dev order, and what will it cost before tax?

3. Temperature

Watch out. In $C = (F - 32) \div 1.8$, subtract 32 **first**. On the TI-30XIIS use brackets: [(] 104 [−] 32 [)] [÷] 1.8 [ENTER]. For a negative number use [(−)], not the minus key.

Example 4

(a) The forecast in Phoenix, Arizona, is a high of 104 °F. What is that in Celsius?

(b) It is −15 °C in Calgary. What is that in Fahrenheit?

(c) Use the quick estimate to change 22 °C to Fahrenheit. Then find the exact answer. Is the estimate good enough to decide what to wear?

(d) A cookie recipe from a US website says to bake at 375 °F. What should you set a Celsius oven to?

4. Mass

Example 5

(a) Fill in the blanks.

3.6 kg = _____ g
750 mg = _____ g
4.2 t = _____ kg

(b) Fill in the blanks.

3 lb 6 oz = _____ oz
74 oz = _____ lb _____ oz

(c) Baby Amira weighed 7 lb 9 oz at birth and 11 lb 4 oz at two months. How much did she gain?

Example 6

Use 1 lb ≈ 0.454 kg, 1 kg ≈ 2.2 lb, 1 oz ≈ 28.35 g and 1 ton ≈ 0.907 t.

(a) An airline allows a checked bag of up to 23 kg. Jordan's suitcase weighs 48 lb on his bathroom scale. Is it under the limit?

(b) Ground beef is $5.99/lb at one store and $12.99/kg at another. Which is the better price? (Use 1 kg ≈ 2.2 lb.)

(c) A bag of chips is labelled 12 oz. What is that in grams?

(d) A load of hay is 1.5 tons. What is that in tonnes?

5. Bushels and Tonnes

Watch out. A tonne holds *many* bushels, so the number of bushels is always bigger than the number of tonnes. If your tonnes answer is bigger, you multiplied when you should have divided.

Example 7

A farmer near Lethbridge harvests 8820 bushels of canola. For canola, use 44.1 bu/t.

(a) How many tonnes of canola is that?

(b) The elevator pays $680.00 per tonne. How much is the crop worth?

(c) What is the price **per bushel**, to the nearest cent?

(d) The farmer also has 92 t of barley. For barley, use 45.9 bu/t. How many bushels is that?